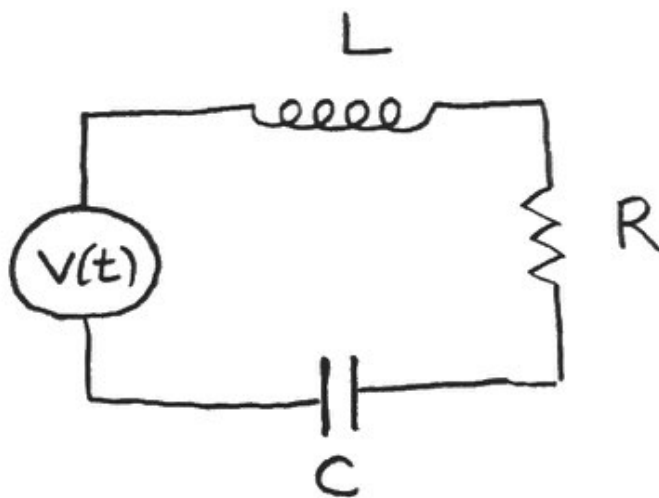


LRC circuits

Consider a circuit with an inductor (L), a resistor (R), a capacitor (C), and a voltage source ($V(t)$).



The voltage across each component is the following.

$$\begin{array}{c} L \\ \text{-----} \\ \text{-----} \end{array}$$
$$V_L = L \frac{di}{dt}$$

$$\begin{array}{c} R \\ \text{-----} \\ \text{-----} \end{array}$$
$$V_R = Ri$$

$$\begin{array}{c} C \\ \text{-----} \\ \text{-----} \end{array}$$
$$V_C = \frac{q}{C}$$

$q = q(t)$ = charge on the capacitor.

$i = i(t) = \frac{dq}{dt}$ = current.

The sum of the voltage across each component of the circuit is equal to the source voltage.

$$\Rightarrow L \frac{di}{dt} + Ri + \frac{q}{C} = V(t)$$

$$\Rightarrow \boxed{L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q(t) = V(t)}$$

LRC circuit DE

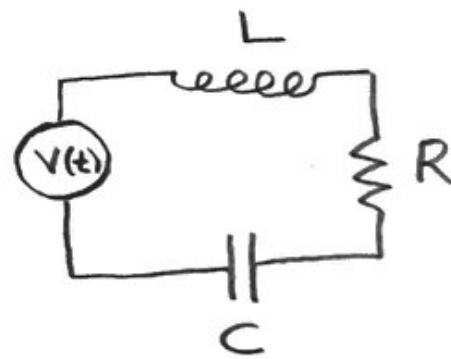
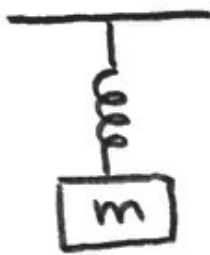
$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q(t) = V(t)$$

mass-spring DE

$$m x''(t) + \beta x'(t) + K x(t) = F_e(t)$$

Observe that two DEs are exactly the same. Everything that we did for mass-spring systems carries over to LRC-circuits.

Concepts for mass-spring systems have a corresponding concept for LRC-circuits.



mass (m)	inductance (L)
Damping Constant (β)	resistance (R)
Spring Constant (k)	$1/\text{capacitance}$ ($1/C$)
External force $F_e(t)$	Voltage source $V(t)$
Position $x(t)$	charge $q(t)$
Velocity $V(t)$	current $i(t)$

A spring behaves like a Capacitor.

A spring can be compressed to store energy.

A capacitor can be charged to store energy.

ex: Consider an LRC-circuit

with $L = 1 \text{ H}$, $R = 100 \Omega$,

$C = 0.0004 \text{ F}$, and $V(t) = 30 \text{ V}$.

(these numbers are chosen for simplicity.)

Find the current $i(t)$ if initially the current is 2 A and the capacitor is uncharged.

$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = V(t)$$

$$\Rightarrow \frac{d^2 q}{dt^2} + 100 \frac{dq}{dt} + 2500 q = 30$$

The initial conditions are:

$$i(0) = 2 \text{ A} \quad \text{and} \quad q(0) = 0.$$

Step ① Find q_c

Roots : $m^2 + 100m + 2500 = 0$

$$(m + 50)^2 = 0$$

$$m = -50$$

$$\Rightarrow q_c = (c_1 + c_2 t) e^{-50t}$$

Step ② Find q_p

Look for $q_p = A$. $[v(t) = 30]$

$$\Rightarrow \frac{dq_p}{dt} = 0 \quad \text{and} \quad \frac{d^2 q_p}{dt^2} = 0$$

$$\Rightarrow 0 + 0 + 2500A = 30$$

$$\Rightarrow A = \frac{3}{250}$$

$$\Rightarrow q_p = \frac{3}{250}$$

Step ③ Find c_1, c_2

The general solution is

$$q(t) = q_c(t) + q_p(t)$$

$$q(t) = (c_1 + c_2 t) e^{-50t} + \frac{3}{250}$$

$$i(t) = \frac{dq}{dt} = c_2 e^{-50t} - 50(c_1 + c_2 t) e^{-50t}$$

$$q(0) = 0 \rightarrow c_1 + \frac{3}{250} = 0$$

$$\rightarrow c_1 = -\frac{3}{250}$$

$$i(0) = 2 \rightarrow c_2 - 50c_1 = 2$$

$$\rightarrow c_2 = 2 + 50c_1$$

$$\rightarrow c_2 = 2 + 50\left(-\frac{3}{250}\right) = \frac{7}{5}$$

charge

$$q(t) = \left(-\frac{3}{250} + \frac{7}{5}t \right) e^{-50t} + \frac{3}{250}$$

current

$$i(t) = \frac{dq}{dt}$$

$$= \frac{7}{5} e^{-50t} + (-50) \left(-\frac{3}{250} + \frac{7}{5}t \right) e^{-50t}$$

$$= \left(\frac{7}{5} + \frac{3}{5} - 70t \right) e^{-50t}$$

$$i(t) = (2 - 70t) e^{-50t}$$